INCH-POUND

MIL-C-17596F
19 August 1988
SUPERSEDING
MIL-C-17596E
25 March 1982

MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING OR ROTARY POWER DRIVEN (EMD) AIR, BASE MOUNTED, 10 HP TO 300 HP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stationary, base mounted, electric motor driven, reciprocating and rotary air compressors that operate in a power range of 10 horsepower (hp) to 300 hp.

1.2 Classification. The compressor unit shall be of the types, styles, classes, groups, drives, and sizes (motor hp), as specified herein. The rated capacity in cubic feet per minute (cfm) and rated discharge pressure in pound-force per square inch gauge (psig) shall be as specified (see 6.2.1).

Type I - Reciprocating (single acting).
Type II - Reciprocating (double acting).
Type III - Rotary (screw).
Type IV - Rotary (sliding vane).

Style A - Base mounted.
Style B - Skid mounted.
Style C - Tank mounted.

Class 1 - With horizontal air receiver.
Class 2 - With vertical air receiver.
Class 3 - Without air receiver.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4310

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Group a - With air-cooled aftercooler.
Group b - With water-cooled aftercooler.
Group c - Without aftercooler.

Drive 1 - Multiple V-belt.
Drive 2 - Direct connected.

Size A	- 10 hp.	Size J	- 75 hp.
Size B	- 15 hp.	Size K	- 100 hp.
Size C	- 20 hp.	Size L	- 125 hp.
Size D	- 25 hp.	Size M	- 150 hp.
Size E	- 30 hp.	Size N	- 200 hp.
Size F	- 40 hp.	Size P	- 250 hp.
Size G	- 50 hp.	Size R	- 300 hp.
Size H	- 60 hp.		

1.3 Definitive part number (DPN). The compressor unit shall be identified by a DPN. This part number is intended for cataloging (see 6.6), and ordering purposes (see 6.2.1). The DPN shall be designated as follows:

DPN designation ---------- M 17596 - X X X X X X

M prefix - items defined by inch-pounds (U.S.) units
Specification number
DPN code for type
DPN code for style
DPN code for class
DPN code for group
DPN code for drive
DPN code for size

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

F-F-351 - Filter and Filter Elements, Fluid, Pressure, Lubricating Oil, Bypass and Full Flow.

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MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704 - Treatment and Painting of Material.
MIL-C-3600 - Compressor, Air and Gas (Except Oxygen) Packaging of.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MS35875 - Air Cleaner, Intake Oil Bath Type, for Industrial Engines.
MS39247 - Air Cleaner, Intake, Dry Type, for Internal Combustion Engines (class I and II).

2.1.2 Other Government documents. The following Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910.95 - Occupational Noise Exposure.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B15.1 - Mechanical Power Transmission Apparatus, Safety Standard for.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Section 8-D-1 - Rules for Construction of Pressure Vessels, Division I.
Power Test Code PTC 9 - Displacement Compressors, Vacuum Pumps and Blowers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 2 - Industrial Control Devices, Controllers and Assemblies, Standards for.
ICS 6 - Industrial Controls and Systems, Enclosures for.
MG 1 - Motors and Generators.
MG 10 - Energy Guide for Selection and Use of Polyphase Motors.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to Rubber Manufacturers Association, 1400 St., NW., Washington, DC 20005).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings.
SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation of Vehicles and Devices.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The compressor unit or unit, as referred to in this specification, consist primarily of sub-assemblies such as the compressing unit (compressor), prime mover (electric motor), instruments and control panel, and if applicable, include the intercooler, aftercooler, air receiver, housing and the required type of mounting. The unit also include components such as safety devices, wirings, and interconnected piping.

3.2 Standard commercial product. The compressor unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the compressor unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 First article. When specified in the contract or purchase order (see 6.2.1), a complete unit shall be subjected to first article inspection.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.6 Design and construction. Unless specified (see 6.2.1), to be packaged for field assembly, all sub-assemblies and components shall be completely assembled and unitized to the required mounting to form a single unit ready for service upon hook-up to field power supply and distribution piping. The unit shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Rotating parts or other parts subject to high temperatures that could be hazardous to operating personnel shall be fully enclosed or provided with protective guards or shields.

3.6.1 Noise level. The compressor noise level (dBA) shall not exceed the value for an 8-hour exposure as defined in OSHA 29 CFR 1910.95 or not to exceed the noise level as specified (see 6.2.1). Any deviation from the OSHA value or to the specified noise level shall be submitted to the contracting officer for approval.

3.7 Ease of maintenance. The compressor and components shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the unit. Ease of maintenance provisions shall insure operation clearances for facilitating maintenance and servicing. Covers or plates that must be removed for component adjustment or for removable parts shall be equipped with substantial boltings or quick-disconnect fastenings.

3.8 Performance. The compressor unit shall be capable of compressing and delivering not less than the specified volume flow rate of air, compressed and delivered at the specified pressure, under atmospheric conditions of 14.7 pound-force per square inch absolute (psia) pressure, 60 degrees Fahrenheit (°F) temperature, and 36 percent relative humidity. In addition, the power required by the compressor shall be calculated in accordance with ASME PTC 9 and shall be the rating indicated by the manufacture with $\pm$3 percent.

3.9 Compressor. The compressor unit shall be capable of continuous operation at rated discharge pressure and capacity. The compressor shall be designed and constructed to withstand the normal strains, jars, vibrations, and other conditions incident to shipping, storage, and installation, so that no parts will work loose in service. All parts subject to wear or service shall be field replaceable, and the compressor shall require only minimum maintenance. Cover plates for access openings shall be oil and dust tight. The cylinder cooling area shall be adequate to prevent excessive heating. Air passages shall be unrestricted. The crankcase, oil reservoir, or receiver shall be provided with means of filling and draining oil and for checking the oil level. On two-stage reciprocating compressors, an intercooler shall be provided to cool the air between stages of compression (see 3.13).

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3.9.1 Type I compressor. The type I compressor shall be a single-acting, reciprocating type, with one or two stages as specified (see 6.2.1). The air temperature at the cylinder discharge flange shall not exceed 380°F, limited by a high temperature shut-off switch. The compressor shall have air-cooled cylinder heads unless water jacketed cooling is specified (see 6.2.1). Cylinder arrangement shall be vertical, radial, or V-type. When compressor is rated at 40 hp or greater, the cylinders shall be separate and individually detachable to make repairs simple and inexpensive. Pistons shall be of the automotive type. Lubrication shall be the splash type or forced feed system (see 3.10, 3.10.1, and 3.10.2).

3.9.2 Type II compressor. The type II compressor shall be a reciprocating, double-acting, crosshead type, with water-cooled cylinders and heads. The compressor shall be single or two-stage as specified (see 6.2.1). The cylinder arrangement may be either horizontal, vertical, V-type, L-type, radial or semi-radial. Cylinder walls shall be of sufficient thickness to permit resleeving or reboring to a radial thickness of at least 1/16 inch, or shall have removable, renewable cylinder liners. Pistons shall have at least two rings and be of minimum weight. Wearing surfaces on the crosshead shall be babbitted, adjustable, or replaceable unless similar surfaces are provided in the crosshead guides. Cylinder stuffing boxes shall be of sufficient depth to assure tightness against compression leakage without excessive piston rod pressure. Oil wiper rings at the frame end shall limit oil loss along the piston rod. The compression cylinder shall be the lubricated type unless oil-free air is specified. When oil-free air delivery is specified (see 6.2.1), no part of the piston rod shall enter both the compression cylinder and the frame end, the piston rings shall be Teflon, and positive means, such as an oil deflection collar, shall be provided to ensure that no oil enters the cylinder via the piston rod. Lubrication shall be as required (see 3.10 and 3.10.2) except the lubrication of horizontal compressors may be the splash type (see 3.10.1). Cylinder lubrication for lubricated type II compressors shall be as specified in 3.10.3.

3.9.3. Type III and type IV compressors. The type III compressor shall be a asymmetrical rotary screw and type IV is a rotary sliding vane compressor. The compressor shall be lubricated and cooled by the injection of filtered lubricating oil into the compression chamber. When the compressor is operated continuously at rated capacity, the temperature of the discharge air shall be not more than 100°F above the ambient air temperature. The compressor shall be equipped with an air/oil separator and receiver. The separator may be located within the receiver. The air/oil receiver shall be designed and constructed in accordance with ASME section 8-D-1 and shall be stamped with the official ASME code symbol. A spring loaded safety valve marked with the ASME Code symbol shall be installed on the receiver. The receiver discharge system shall include an automatic blowdown valve which shall reduce the pressure in the receiver to approximately 12 psig by discharging air to atmosphere when the compressor is shutdown. A check valve shall be installed to prevent the backflow of system air into the air/oil receiver when the compressor is shutdown. The compressor shall automatically shutdown if the discharge air temperature exceeds the limits of the compressor's normal

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operating range. Compressors rated at 40 hp or greater shall operate at 30 percent of rated hp or less when operating unloaded. Lubrication shall be in accordance with 3.10 and 3.10.4.

3.9.4 Bearings. Main bearings and connecting rod bearings shall be of the type ordinarily employed by the manufacturer as his standard commercial practice.

3.9.5 Valves. Valves for all reciprocating compressors shall operate quietly and shall be readily removable for inspection.

3.9.6 Air intake. The air intake shall be suitably located to avoid the exhaust heat resulting from the compressor's cooling system. Airtight inlet piping shall be capable of resisting corrosion. The piping shall be free of sharp bends and shall be suitably routed so as not to interfere with the function and maintenance of adjacent components.

3.9.7 Air filter. The compressor air intake shall be fitted with an air filter-silencer in accordance with the manufacturer's latest standard practice. The air filter shall act as a muffler of the intake noises and shall have minimum restriction of air flow. The filter shall be of the oil-wetted type or dry type of equal efficiency and shall be readily removable for cleaning. When specified (see 6.2.1), the air filters shall be in accordance with either MS39247 or MS35875.

3.9.8 Safety valves. When no receiver is supplied, a safety valve shall be installed in the discharge piping of all compressors. A safety valve shall be furnished between compressor stages. When the receiver is supplied, a safety valve shall be installed on the receiver. The safety valve(s) shall be the spring-loaded type, marked to show its capacity and pressure rating, and stamped with the ASME Code symbol. Both the safety valve(s) and its method of installation shall be in accordance with ASME section 8-D-1.

3.9.9 Discharge shut-off valve. When specified (see 6.2.1), the compressor discharge shall be equipped with the shutoff valve. The shutoff valve shall be installed downstream of the safety valve.

3.10 Frame lubrication. The lubrication system for each type of compressor shall include an oil level indicating device and an oil drain with manufacturer's standard drain plug. The oil filter, if required (see 3.10.2 and 3.10.4), shall be in accordance with the manufacturer's standard practice, except, when specified (see 6.2.1), the oil filter shall be either of the bypass or fullflow type in accordance with F-F-351. When specified (see 6.2.1), the oil receiver, crankcase, or reservoir shall be equipped with an electric, thermostatically controlled immersion heater.

3.10.1 Splash type lubrication. The splash type lubrication system shall have an oil reservoir or crankcase from which the compressor running gear is lubricated. The motor shall automatically shut off if the compressor oil level gets too low.

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3.10.2 Forced feed lubrication. The forced feed lubrication system for reciprocating compressor running gear shall include an oil reservoir or crankcase, a positive displacement pump, an oil filter, and a low oil pressure device for automatic shut off of the motor. The positive displacement pump shall incorporate a pressure relief device to prevent excessive oil pressure from building up downstream of the pump. An oil cooler in accordance with the manufacturer's standard design shall be provided if required for the type of compressor specified.

3.10.3 Cylinder lubrication. Cylinder lubrication for lubricated type II compressors shall be provided by means of a forced-fed lubricator operated from the compressor running gear. The oil feed to each cylinder shall be visual and the oil feed regulation shall be adjustable.

3.10.4 Rotary compressor lubrication. The lubrication system for rotary compressors, shall include an oil separator, receiver, cooler, and filter. The oil separator shall reduce the level of oil entrained in the discharge air to 10 parts per million or less. When a pressure in excess of 12 psi must be maintained in the receiver to ensure adequate circulation of compressor oil or to eliminate any carry-over, the air discharge piping for the receiver shall include a minimum pressure valve set to maintain a predetermined minimum back pressure on the compressor oil system. The oil cooler shall either be the water-cooled shell and tube type or the air-cooled fin and tube type. The oil temperature of a shell and tube type oil cooler shall be controlled by a thermostat that automatically regulates the flow of water or oil through the cooler. The lubrication system shall include a device for automatically stopping the motor if the oil level or oil pressure becomes low. Check valves or other means shall be provided to prevent loss of oil through the intake filter.

3.11 Regulation. Regulation of the compressor shall be accomplished by one of the following two methods of control as specified (see 6.2.1). The regulation shall have adjustments for changing discharge and differential pressures.

3.11.1 Constant speed control. The compressor shall operate continuously at constant speed. Means shall be provided to automatically load and unload the compressor at preset minimum and maximum pressure settings respectively. Loading and unloading on two-stage double acting compressors shall be accomplished by a minimum of two steps (full load and no load). Means shall be provided for the automatic release of pressure within the cylinders after shutdown or when unit is operating without load. Means shall also be provided for the manual or automatic unloaded starting of the unit. When specified (see 6.2.1), the compressor shall be equipped with an adjustable timed control to shutdown the motor after operating continuously unloaded for periods of up to 30 minutes.

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3.11.2 Dual control. The dual system shall consist of a combination of constant speed control and an automatic start/stop control by automatic or manual selector switch. When set for start/stop control, the motor shall stop automatically when the discharge pressure reaches the maximum pressure setting and start automatically when the discharge pressure falls to the minimum setting. The compressor shall unload immediately after shutdown and prior to restarting and the cooling water, when applicable, shall shut off automatically during periods of motor shutdown.

3.12 Optional safety controls. In addition to controls specified herein, the compressors shall be equipped with the optional safety controls listed in table I as specified (see 6.2.1). The controls shall provide an alarm of the appropriate condition by sounding a horn and by flashing an amber light or shall cause a shutdown of the compressor as indicated in the table for each type of compressor. Gages and visual identification lights for control points shall be panel mounted or mounted on a separate working height console as specified (see 6.2.1).

TABLE I. Optional safety controls.

Control point	Type I		Type II		Type III and Type IV	
	Alarm	Shut Down	Alarm	Shut Down	Alarm	Shut Down
High-lubrication oil temp	–	–	x	x	x	x
Cylinder lubrication failure	–	–	–	x	–	–
High main bearing temperature	–	x	–	x	–	–
High discharge air temperature	x	x	x	x	x	x
High discharge air pressure	x	x	x	x	x	x
High intercooler air temperature	x	x	x	x	–	–
Low cooling water flow	–	x	–	x	–	–
Low water pressure	–	x	–	x	–	–
High water temperature	x	x	x	x	–	–
High intercooler moisture separator level (two-stage compressors only)	x	x	x	x	–	–
Excessively high motor temp.	x	–	x	–	x	–
Excessive vibration	–	x	–	x	–	–

3.13 Intercoolers and aftercoolers. Intercoolers shall be supplied on all two stage compressors. Intercoolers for air-cooled compressors shall be the tube and fin type. Intercoolers for water-cooled compressors shall be the shell and tube type, except that tube and fin type may be used when the intercooler is supported by the compressor frame or attached to the compressor. Air or water cooled intercoolers may be the integral cast type when compressor hp is 25 hp or less. Aftercoolers shall be of the water-cooled shell and tube type or air-cooled tube and fin type. Water cooled aftercoolers and intercoolers shall be of sufficient capacity to cool

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the compressed air to not less than 15°F and 20°F, respectively, of the temperature of the water entering the cooler. Air-cooled intercoolers and aftercoolers shall have sufficient capacity to cool the compressed air to within 20°F of the ambient temperature under the atmospheric conditions specified in 3.8. Water-cooled intercoolers and aftercoolers shall be provided with sight-flow indicator to visually observe the flow of water to the cooler. The pressure drop of compressed air through the cooler shall not exceed 1 psi for operating pressures above 70 psi or 2.5 psi for operating pressures equal to or below 70 psi. All intercoolers and aftercollers shall be provided with a moisture separator and automatic drain trap to remove the condensed moisture and oil from the air leaving the cooler.

3.13.1 Shell and tube. Shell and tube coolers shall be of the floating head type consisting of a removable and cleanable nest of corrosion resistant tubes within a steel shell. Air may pass either through the tubes or the shell.

3.13.2 Tube and fin. Tube and fin coolers shall be copper, aluminum, copper-aluminum, or copper-alloy construction. Fins shall be securely bonded to the tubing. Tube and fin coolers shall be provided with a fan for circulation of the cooling air. The fan shall be adequately guarded for safety and be driven either from the compressor crankshaft or by an independent electric motor.

3.14 Gages. Dial type pressure indicating gages shall be provided with all compressors to indicate discharge or receiver air pressure, air pressure between stages for two-stage compressors 25 hp and larger, and oil pressure for pressurized or force feed lubrication system. Gages shall have an approximate range of twice the rated pressure or normal operating pressures.

3.15 Air receiver. Unless otherwise specified (see 6.2.1), the receiver volume shall be the manufacturers calculated volume required for delivery of non-pulsating flow of compressed air at the rated discharge pressure. The receiver shall be designed and constructed in accordance with ASME section 8-D-1 and shall be stamped with the official code symbol. The receiver shall be designed for a working pressure not less than 10 percent greater than the maximum working pressure of the compressor cylinder. The receiver shall be equipped with a safety valve (see 3.9.8) and a drain valve.

3.16 Housing. When specified (see 6.2.1), the rotary compressor shall be provided with a sheet metal housing which shall enclose the motor, compressor, and instrument panel. The housing shall include quick opening side panels or hinged access openings to the motor, compressor, and instrument panel. The housing shall include grills that permit maximum passage of cooling air for the air-cooled radiator and oil cooler of the compressor. Sheet metal for the housing and panels shall be steel not less than 0.0598-inch (US revised standard gage No. 16) thick. Supporting members and braces shall be furnished as required to insure rigidity and to minimize vibrational noise during operation. Panels shall be sized to provide maximum accessibility to the compressor components. Upward-opening hinged access panels shall be equipped with quick-disconnect fasteners. Housing shall be sound attenuated to limit the noise level of the unit (see 3.6.1).

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3.17 Motors. The compressor motor shall conform to NEMA MG-1 and shall be suitable for continuous duty. The motor shall be designed for operation with electrical power supply having characteristics as specified (see 6.2.1). The motor shall have a hp nameplate rating equal to or greater than 110 percent of the hp required for continuous operation of the compressor at full load. Dripproof motors shall be poly-phase induction type. When specified (see 6.2.1), motors 125 hp and larger shall be the synchronous type. Motors that drive auxiliary components shall conform to NEMA MG-1 and shall have electrical power supply characteristics as specified (see 6.2.1).

3.17.1 Motor efficiency. Motor efficiency shall be in accordance with the criteria of NEMA MG-10 and efficiency tables of NEMA MG-1. When specified (see 6.2.1), a high efficiency motor shall be furnished.

3.18 Motor starter. A magnetic starter conforming to NEMA ICS-2 shall be furnished with capacity and electrical characteristics suitable for the motor in accordance with NFPA 70. When specified (see 6.2.1), motors rated above 50 hp shall have reduced voltage, partial winding, or other means to limit starting current inrush to 200 percent of the normal full load value. The starter shall include thermal overload protection and type 1 enclosure in accordance with NEMA ICS-6.

3.18.1 Starting control circuit. Unless otherwise specified (see 6.2.1), the control circuits for motors shall be nominal 110 Volts alternating current. When specified (see 6.2.1), a spare thermal overload relay shall be furnished and securely fastened inside the starter enclosure.

3.19 Drive. When reciprocating compressor is connected to a synchronous motor, a direct drive shall be provided. Direct drive units shall either be of the close-coupled, integral-coupled, or flexible-coupled type as specified (see 6.2.1). Flexible couplings shall be guarded. Couplings for direct drive compressors shall be designed to operate safely at 150 percent of the motor hp rating. Drivers and belts for belt driven compressors shall conform to RMA Standard. Belts shall be designed with a service factor of at least 1.5. Belt-driven units shall have means for adjusting belt tension. Belt guards shall conform to ANSI B15.1.

3.20 Electromagnetic interference characteristics. When specified (see 6.2.1), the compressor shall conform to the electromagnetic interference suppression requirements and test limits of SAE J551.

3.21 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.22 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2.1), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as soil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.23 Lubricants. Lubricants shall be in accordance with the equipment manufacturer's recommendations and shall be compatible with lubricants specified herein. Pressure lubrication shall not damage seals or other parts. Unless specified otherwise (see 6.2.1), all parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used. The equipment shall be designed to operate efficiently and satisfactorily when lubricated with standard military lubricants (see 6.4).

3.23.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534 and shall be located in accessible, protected positions. A bright red circle shall be painted around each lubricating point.

3.24 Mounting. The required mounting shall be of sufficient size, strength, and rigidity to facilitate the installation and assembly of all the required sub-assemblies and components.

3.24.1 Skid mounted. The skid shall consist of two parallel steel skid runners with an integrally formed, welded, or bolted base plate for mounting components of the unit. The skid runners shall be upturned or beveled at each end to permit skidding the unit into place. The skid runners shall be furnished with mounting holes for anchoring the unit to a concrete foundation or wooden deck. The number and size of holes required shall be as determined by the manufacturer for anchorage of the unit under the most severe condition of vibration or thrust likely to develop during operation. Individual mountings shall be furnished on the base plate as required for supporting motor, compressor, and other components. Other components requiring removal for service, adjustment, or horizontal alignment shall be mounted in a manner to permit such removal or alignment. Each end of the skid shall be furnished with pulling attachment designed to be used also as a lifting and tiedown device.

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3.24.2 Base mounted. The baseplate shall be a single fabricated steel unit provided with access openings for grouting the load carrying structural members for installation on a concrete foundation. When specified (see 6.2.1), the baseplate main frame shall be provided with openings designed for a forklift lifting of the whole compressor unit.

3.24.3 Tank mounted. The tank (horizontal or vertical) shall be provided with bottom supports for bolting to a concrete foundation.

3.25 Lifting and tiedown attachments. When specified (6.2.1), the compressor shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the compressor. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the compressor on the carrier when shipped.

3.26 Instruction plates. The compressor unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.27 Identification marking. Identification shall be permanently and legibly marked directly on the compressor and sub-assemblies or on a corrosion-resisting metal plate securely attached to the compressor and sub-assemblies at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.27.1 Rotation arrow. Rotation arrow shall be cast-in or attached near the drive end of the compressor. If attached, the rotation arrow and attachment pins shall be of corrosion resisting materials.

3.28 Spare parts and maintenance tools. When specified (see 6.2.1), spare parts and maintenance tools shall be furnished. Special tools shall be kept to a minimum.

3.28.1 Toolbox. When specified (see 6.2.1), a toolbox shall be provided. The toolbox shall be large enough to store specialized tools not carried in mechanics toolbox, and required for field service or maintenance, but shall not have external closed dimensions less than 14 inches in length, 6 inches in width and 6 inches in height. The toolbox shall be nominal 0.0747-inch (U.S. revised standard gage No. 14) thick steel with a hinged lid and a trunk drawbolt to keep the lid secured. The toolbox shall be mounted in a protected, accessible location.

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3.29 Workmanship.

3.29.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.29.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.29.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.29.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.29.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification

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shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article inspection shall be performed on one compressor unit when a first article is required (see 3.4 and 6.3). This inspection shall include the examination of 4.5, the tests of 4.6, and, when required (see 4.7), the preproduction pack inspection. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on the compressors. This inspection shall include the examination of 4.5, the tests of 4.6.1, 4.6.2, and 4.6.4, and the packaging inspection of 4.7.

4.5 Examination. Each compressor unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. Each compressor shall be tested, and any unit failing to pass the following tests, as applicable, shall constitute cause for rejection. Certified test reports from an independent laboratory approved by the contracting officer will be accepted as evidence that the unit has passed the required tests. Tests shall be conducted as outlined in the referenced documents as herein specified. Prior to the tests specified hererin which require the compressor(s) to be operated, the unit(s) shall be serviced with the oils and grease recommended by the manufacturer for use in the ambient temperature at the test location.

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4.6.1 Receiver test. The receiver shall be hydrostatically tested in accordance with the requirements of ASME section 8-D-1 and held at this pressure until examined for leaks and distortion. Leaks or distortion shall constitute failure of this test. The ASME Code symbol stamp will be accepted as evidence of this test.

4.6.2 Safety valve test. The safety valve shall be tested to determine if its relieving capacity and popping pressure are in accordance with requirements of ASME section 8-D-1.

4.6.3 Performance test. The compressor shall be tested to determine capacity and power consumption in accordance with the rules, procedures, and instrumentation specified in ASME PTC 9. The duration of the test shall be sufficient to record all data required under Report of Test of PTC 9, but in no case shall the test duration be less than 1 hour. Test results shall be corrected to the atmospheric conditions (see 3.8). Prior to the actual performance test, preliminary tests shall be run, as required, to verify that the compressor and associated systems are in suitable condition, that regulation controls have been properly adjusted, and that gages and instruments are functioning properly. Inability of the compressor to meet the performance requirements of the specification or to operate satisfactory shall constitute failure of this test.

4.6.4 Operational test. Each compressor, excluding the unit offered for first article inspection, shall be tested for not less than 15 minutes to verify satisfactory operation of the compressor and all controls. During the test, all necessary adjustments in controls shall be made as required to insure that performance requirements relating to the compressor output and regulation shall be met.

4.6.5 Test for electromagnetic interference characteristics. When electromagnetic interference characteristics are required, the compressor shall be tested to determine conformance to SAE J551. In lieu of tests, the manufacturer may furnish a certification that the compressor meets the requirements, together with a list of the suppression devices installed. The list shall be sufficietly detailed to allow visual determination that the devices are installed.

4.7 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of Section 4 of MIL-C-3600. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a preproduction pack shall be furnished for examination and test within the time frame required.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-C-3600 with the level of preservation and the level of packing as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. This specification covers air compressors intended for stationary service in installations where a steady volume of air in a pressure range of 5 to 125 psi is required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Rated capacity (cfm) and rated discharge pressure (psig) required (see 1.2).
c. DPN designation (see 1.3).
d. When first article is required (see 3.4).
e. When sub-assemblies and components are to be packaged for field assembly (see 3.6).
f. Specified maximum noise level (dBA) if other than OSHA recommended value (see 3.6.1).
g. Number of stages required (see 3.9.1 and 3.9.2).
h. When water jacketed cooling is required for type I compressor (see 3.9.1).
i. When oil free delivery is required for type II compressor (see 3.9.2).
j. When air filter shall be in accordance with MS39247 or MS35875 (see 3.9.7).
k. When a shutoff valve is required in compressor discharge (see 3.9.9).
l. When oil filter is to be a full flow or bypass type in accordance with F-F-351 (see 3.10).
m. When an electrically controlled immersion heater is required (see 3.10).
n. Required method of compressor regulation (see 3.11).
o. When an adjustable timed control is required (see 3.11.1).
p. When optional safety controls are required (see 3.12).
q. When gages and identification lights are panel mounted or mounted on a separate console (see 3.12).
r. Required receiver volume if other than manufacturers calculated volume (see 3.15).
s. When housing or enclosure is required (see 3.16).
t. Required electrical power supply characteristics for compressor motor (see 3.17).
u. Required electrical power supply characteristics for auxiliary motor (see 3.17).
v. When motors 125 hp and larger shall be synchronous motor (see 3.17).
w. When high efficiency motor is required (see 3.17.1).
x. When reduce voltage, partial winding or other means of limiting starting current is required (see 3.18).
y. Required electrical power supply characteristics for control circuits if other than specified (see 3.18.1).

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z. When a spare overload relay shall be furnished (see 3.18.1).
aa. When direct drive is to be a close-coupled, integral-coupled or flexible-coupled (see 3.19).
bb. When electromagnetic interference characteristic is required (see 3.20).
cc. When fungus resistance is required (see 3.21).
dd. When MIL-T-704, type A, treatment and cleaning is required (see 3.22).
ee. Color of finished coat required (see 3.22).
ff. When initial lubrication shall be other than specified (see 3.23).
gg. When baseplate main frame shall be provided with openings for forklift lifting (see 3.24.2).
hh. When lifting and tiedown attachment is required (see 3.25).
ii. When spare parts and maintenance tools shall be furnished (see 3.28).
jj. When toolbox is required (see 3.28.1).
kk. When a preproduction pack is required and time frame for submission (see 4.7).
ll. Level of preservation and level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Standard lubricants. When requested by the contractor, the contracting officer should furnish a list of standard military lubricants applicable to the equipment covered by this specification. Standard military lubricants are listed in the DoD section of the Federal Supply Catalog C9100-IL.

6.5 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.6 Cataloging data. For cataloging data purposes, DPN code numbers are assigned to type, style, class, group, drive, and size.

a. DPN code for type
1 = Type I
2 = Type II
3 = Type III
4 = Type IV

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b. DPN code for style
A = Style A
B = Style B
C = Style C

c. DPN code for class
1 = Class 1
2 = Class 2
3 = Class 3

d. DPN code for group
a = Group a
b = Group b
c = Group c

e. DPN code for drive
1 = Drive 1
2 = Drive 2

f. DPN code for size

A = Size A	J = Size J
B = Size B	K = Size K
C = Size C	L = Size L
D = Size D	M = Size M
E = Size E	N = Size N
F = Size F	P = Size P
G = Size G	R = Size R
H = Size H	

6.7 Example of DPN. Requirements: 200 horsepower, double acting reciprocating compressor, base mounted, furnished with vertical air receiver and water-cooled aftercooler, direct drive.

DPN designation: M17596 - 2A2b2N

6.8 Cross reference of classification. The classification of the compressor unit in this document differs from the superseded specification in the following respect:

MIL-C-17596E	MIL-C-17596F
Type I	Type I
Type II	Type II
Type III	Type III
not designated	Type IV
not designated	Style A
not designated	Style B
not designated	Style C

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not designated	Class 1
not designated	Class 2
not designated	Class 3
not designated	Group a
not designated	Group b
not designated	Group c
not designated	Drive 1
not designated	Drive 2
not designated	Size A
not designated	Size B
not designated	Size C
not designated	Size D
not designated	Size E
not designated	Size F
not designated	Size G
not designated	Size H
not designated	Size J
not designated	Size K
not designated	Size L
not designated	Size M
not designated	Size N
not designated	Size P
not designated	Size R

6.9 Subject term (key word) listing.

Axial screw
Base mounted
Compressor
Compressor, reciprocating
Compressor, rotary
Double acting
Drive, direct connected
Drive, multiple V-belt
Electric motor driven
Single acting
Skid mounted
Sliding vane
Stationary
Tank mounted

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6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activities:
Air Force - 84
DLA - CS

Preparing activity:
Navy - YD

(Project 4310-0172)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-17596F	COMPRESSOS, RECIPROCATING OR ROTATRY POWER DRIVEN (EMD) AIR, BASE MOUNTED, 10 HP TO 300 HP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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